Solutions

Algebraic problems

Question 2.1: Differentiate the following functions with respect to x:

(a) $f(x) = 5x^2 \implies \frac{df}{dx} = 10x$
(b) $g(x) = 10x^3 \implies \frac{dg}{dx} = 30x^2$
(c) $h(x) = 11x^1 \implies \frac{dh}{dx} = 11$

Question 2.2: Differentiate the following polynomials with respect to x:

(a) $f(x) = 5x^3 + 2x^2 + 3x + 5 \implies \frac{df}{dx} = 15x^2 + 4x + 3$
(b) $g(x) = 10x^3 + x + 5 \implies \frac{dg}{dx} = 30x^2 + 1$
(c) $h(x) = 11x^5 + 2x^4 + 2x^3 + x^2 + 3x + 2 \implies \frac{dh}{dx} = 55x^4 + 8x^3 + 6x^2 + 2x + 3$

Question 2.3: Find the gradient of the following functions at the values of x specified:

(a) $f(x) = x^2 - 3x$ at $x = 10 \implies \frac{df}{dx} = 2x - 3, \ \frac{df}{dx}(10) = 17$
(b) $f(x) = 5x^5 - 2x$ at $x = 0 \implies \frac{df}{dx} = 15x^4 - 2, \ \frac{df}{dx}(0) = -2$
(c) $f(x) = 10x - 3x^7$ at $x = -2 \implies \frac{df}{dx} = 10 - 21x^6 \ \frac{df}{dx}(-2) = -1334$

Question 2.4: What function when differentiated gives the following [Hint: you will need to do the process of differentiation in reverse]:

(a) $\frac{dy}{dx} = 2x \implies y = x^2$
(b) $\frac{dy}{dx} = -3x^3 \implies y = -\frac{3}{4}x^4$
(c) $\frac{dy}{dx} = 7 \implies y = 7x + d$

[d is a constant. Why do you think we need this? Ask in class if this is unclear]

Question 2.5: Obtain the derivative of the following functions:

(a) $y = 10x(5x + 3x) = 80x^2 \implies \frac{dy}{dx} = 160x$
(b) $y = 50x^2 + 30x^2 = 80x^2 \implies \frac{dy}{dx} = 160x$
(c) $y = 3ce^{ix} \implies \frac{dy}{dx} = 3cie^{ix}$
(d) $y = \sqrt{x} = x^{\frac{1}{2}} \implies \frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} = 1/2\sqrt{x}$
(e) $y = \cos(x) \implies \frac{dy}{dx} = -\sin(x)$
(f) $y = e^{5x} \implies \frac{dy}{dx} = 5e^{5x}$
(g) $y = 3x + 5y \implies y = -\frac{3x}{4} \implies \frac{dy}{dx} = -\frac{3}{4}$
(h) $y = 2ln(x) \implies \frac{dy}{dx} = 2/x$

Question 2.6: Obtain the derivative of the following (slightly harder) functions:

(a) $y = \sum_i 3ce^{ix} \implies \frac{dy}{dx} = 3cie^{ix}$ (Remember the differential of a sum is the sum of the

differential. See Section 2.4)

(b) $y = \sqrt{\frac{4x}{\pi}} = \frac{2}{\sqrt{\pi}}\sqrt{x} \implies \frac{dy}{dx} = \frac{2}{\sqrt{\pi}}\frac{1}{2\sqrt{x}} = \sqrt{\frac{1}{\pi x}}$

(c) $y = \cos(x) + \sin(x) \implies \frac{dy}{dx} = -\cos(x) + \sin(x)$

(d) $y = e^{5x+4} \implies \frac{dy}{dx} = 5e^{5x+4}$

(e) $y = ln(2x) = ln(2) + ln(x) \implies \frac{dy}{dx} = \frac{1}{x}$

(f) $y = 4x^4 + e^x + \sin(x) \implies \frac{dy}{dx} = 8x^4 + e^x + \cos(x)$

Question 2.7:

(a) Differentiating Equation 2.2 gives us:

$$N(t) = N_0 e^{-\lambda t} \quad \implies \quad \frac{dN}{dt} = -N_0 \lambda e^{-\lambda t}$$

substituting this in Equation 2.1 gives us

$$-N_0 \lambda e^{-\lambda t} = -\lambda N_0 e^{-\lambda t}$$

Alternatively, arranging the equation we see that:

$$\begin{aligned} \frac{dN}{dt} &= -\lambda \underbrace{N_0 e^{-\lambda t}}_{N} \\ &= -\lambda N \end{aligned}$$

Therefore Equation 2.2 is indeed the solution to the radioactive decay equation!

(b) A radioactive material has a half-life of $t_{1/2} = 30$ years and there are $10,000$ nuclei in the sample, so at 90 years, the number of nuclei is

$$\begin{aligned} N(90) &= N_0 \times 2^{-n} \\ &= 10000 \times 2^{-3} \\ &= \frac{10000}{8} \\ &= 1250 \end{aligned}$$

There are 1250 nuclei remaining after 90 years.

(c) A radioactive material has a half-life of $t_{1/2} = 30$ years we see that the decay parameter is:

$$\begin{aligned} \lambda &= \frac{\ln 2}{t_{1/2}} \\ &= \frac{\ln 2}{30} \end{aligned}$$

So after 90 years we see that

$$\begin{aligned} N(90) &= N_0 \times \exp(\lambda \times 90) \\ &= 10000 \times \exp\left(\frac{\ln 2}{30} \times 90\right) \\ &= 10000 \times \exp(3\ln 2) \\ &= 1250 \end{aligned}$$

There are 1250 nuclei remaining after 90 years (i.e. the same).

Question 2.8: Differentiating $P(z)$ with respect to z gives us:

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right) \quad \Longrightarrow \quad \frac{dP}{dz} = -P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right)$$

substituting these in Equation 2.1 gives us

$$-P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right) = -P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right)$$

Alternatively, arranging the equation we see that:

$$\begin{aligned} \frac{dP}{dz} &= -\frac{g}{R_a T} \underbrace{-P_0 \exp\left(-\frac{g}{R_a T} z\right)}_{P} \\ &= -\frac{P}{R_a T} g \end{aligned}$$

Therefore $P(z)$ is indeed the solution to the hydrostatic equation!

Question 2.9:

(a) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t}{25} \quad \Longrightarrow \quad \frac{dV}{dt} = -\frac{V_0}{25}$$

We can see that the drainage rate is constant here.

(b) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t^2}{625000} \quad \Longrightarrow \quad \frac{dV}{dt} = - - (V_0)\frac{2t}{625000}$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -10^{-2} \times \frac{2 \times 600}{625000} = -1.92 \times 10^{-5}$$

Using the central differences method:

$$V(590) = 10^{-2} - 10^{-2}\frac{590^2}{625000} = 0.00443$$
$$V(610) = 10^{-2} - 10^{-2}\frac{590^2}{625000} = 0.00405$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.00405 - 0.00443}{20} = -1.90 \times 10^{-5}$$

(c) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t^3}{312500000} \quad \Longrightarrow \quad \frac{dV}{dt} = -(V_0)\frac{3t^2}{312500000}$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -10^{-2} \times \frac{3 \times 600^2}{312500000} = -3.46 \times 10^{-5}$$

Using the central differences method:

$$V(590) = 10^{-2} - 10^{-2}\frac{590^3}{312500000} = 0.003428$$
$$V(610) = 10^{-2} - 10^{-2}\frac{590^3}{312500000} = 0.002737$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.002737 - 0.003428}{20} = -3.46 \times 10^{-5}$$

(d) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 \exp\left(-\frac{t}{300}\right) \quad \Longrightarrow \quad \frac{dV}{dt} = -\frac{V_0}{300}\exp\left(-\frac{t}{300}\right)$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -\frac{10^{-2}}{300}\exp\left(-\frac{600}{300}\right) = -4.5 \times 10^{-6}$$

Using the central differences method:

$$V(590) = 10^{-2}\exp\left(-\frac{600}{300}\right) = 0.001399$$
$$V(610) = 10^{-2}\exp\left(-\frac{600}{300}\right) = 0.001309$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.001309 - 0.001399}{20} = -4.5 \times 10^{-6}$$